

TraceScope AI 2.0

Research Upgrade, Verification & Publication Roadmap

Scanner Source Attribution + Document Forensic Anomaly Analysis

Prepared as a phase-by-phase execution and verification guide

September 2026

1. Purpose of This Document

This document is a practical roadmap for upgrading TraceScope AI from an internship-level technical project into a reproducible research system suitable for a research paper. Each phase has a defined objective, implementation work, expected files/results, verification criteria, and a stop/go checkpoint. Do not move to the next phase until the current checkpoint passes.

The central research question is: Can scanner-source attribution be improved by combining noise-residual representations and explicit forensic features such as PRNU correlation, frequency-domain characteristics, texture descriptors, gradients, and statistical measurements?

2. Important Scientific Position

The current documentation reports a Hybrid CNN accuracy above 96.4%, but that number should be treated as a project result until the exact dataset composition, split strategy, device independence, and evaluation procedure are documented and reproduced. For a paper, do not claim that a softmax confidence score is legal proof or that a scanner model prediction necessarily identifies one physical device.

- Distinguish scanner-model classification from individual physical-device attribution.
- Prevent train/test leakage caused by patches from the same source image or acquisition session.
- Use an independent test set and document exactly how it was constructed.
- Report more than accuracy: precision, recall, F1, confusion matrix, and confidence/calibration where appropriate.
- Use ablation studies to show which components actually contribute.
- Use robustness and open-set experiments to test behavior outside ideal conditions.
- Describe tampering detection as anomaly/consistency analysis unless independently validated as a forensic proof mechanism.

3. Current TraceScope Architecture

Your current design contains both traditional machine learning and deep learning:

Component	Type	Input	Role
Random Forest	Traditional ML	10 statistical/metadata features	Baseline classifier
SVM (RBF)	Traditional ML	Scaled feature vector	Baseline classifier
ResNet-18	Deep Learning	High-pass/noise residual patches	Deep scanner classification
Hybrid CNN	Deep Learning	Residual image + 17 handcrafted features	Primary fusion model
PRNU / FFT / LBP / Sobel / statistics	Signal processing / feature extraction	Image/residual	Create forensic descriptors

4. Target TraceScope 2.0 Architecture

The target system should be organized into the following logical pipeline:

Input image → preprocessing/alignment → residual extraction → PRNU/FFT/LBP/Sobel/statistics → CNN residual branch → feature fusion → scanner attribution → unknown-scanner decision → patch-level anomaly analysis → report.

5. Master Phase Roadmap

Phase	Goal	Main Output	Checkpoint
0	Repository/code audit	Implementation inventory	Every claimed component is verified
1	Dataset engineering	Clean dataset + manifest	Counts and labels are reproducible
2	Leakage-free split	Train/val/test manifests	No acquisition/document leakage
3	Forensic preprocessing	Residual/fingerprint pipeline	Visual + numerical sanity checks pass
4	Baseline ML	RF/SVM benchmark	Baseline metrics reproducible
5	Deep CNN baseline	ResNet experiment	Independent test metrics
6	Hybrid CNN	Fusion model	Hybrid reproducibly evaluated
7	Ablation study	Component contribution table	Each component tested
8	Robustness	Transformation performance matrix	Failure modes documented
9	Open-set attribution	Unknown scanner detector	Known vs unknown tested
10	Tampering/anomaly	Patch heatmap + metrics	Localization evaluated
11	Explainability	Grad-CAM/feature explanation	Qualitative examples
12	Reproducibility	Experiment config + seeds + logs	Run can be repeated
13	Paper	Research manuscript	Claims match evidence

6. Phase 0 — Audit the Existing Repository

Objective: determine what is actually implemented, rather than relying only on the documentation.

6.1 What to inspect

- Open `src/baseline/train_baseline.py`, `predict_baseline.py`, and `evaluate_baseline.py`.
- Open `src/cnn_model/model.py`, `dataset.py`, `train.py`, and `evaluate.py`.
- Open `src/hybrid_cnn/model.py`, `feature_extrac.py`, `processing.py`, `utils.py`, `train_hybrid_cnn.py`, and `eval_hybrid_cnn.py`.
- Inspect `landing_page.py` and verify which models the dashboard can actually load.
- Inspect requirements file and record Python, TensorFlow, PyTorch, OpenCV, scikit-learn and image-processing versions.
- Confirm every saved model artifact exists and can be loaded without retraining.

6.2 Create an implementation inventory

Claim in documentation	Actually implemented?	Evidence/file	Needs work?
Random Forest	YES/NO	<code>src/baseline/...</code>	Record
SVM	YES/NO	<code>src/baseline/...</code>	Record
ResNet-18	YES/NO	<code>src/cnn_model/...</code>	Record
Hybrid CNN	YES/NO	<code>src/hybrid_cnn/...</code>	Record

PRNU reference fingerprints	YES/NO	feature extractor/artifact	Record
Tampering heatmap	YES/NO	dashboard/source	Record
GPU acceleration	YES/NO	processing/utils	Record

6.3 Expected output

Create: docs/implementation_audit.md or .csv containing every component, source file, status, and known issue.

6.4 Verification checkpoint

PASS only when every major claim in the project documentation is mapped to executable code or explicitly marked as future work.

7. Phase 1 — Dataset Engineering

Objective: create a dataset that supports scientifically meaningful scanner attribution.

7.1 Record the right metadata

- scanner_model
- physical_device_id (if multiple physical units are available)
- document_id
- acquisition_session_id
- DPI/resolution
- image_path
- image format
- color/grayscale status
- document type/content category
- date/session identifier where appropriate
- tampered/untampered label if used for anomaly experiments

7.2 Recommended directory design

data/processed/<scanner_model>/<device_id>/<session_id>/<document_id>.<ext>

Keep the original data unchanged. Create processed copies for resizing, residual extraction, and augmentation.

7.3 Critical issue: model vs device

If all Epson V39 samples come from one physical Epson V39, the experiment demonstrates recognition of that acquisition source under your dataset conditions, but it does not prove generalization to every Epson V39.

7.4 Expected output

- dataset_manifest.csv
- class_counts.csv
- device_counts.csv (if physical devices are tracked)
- data_quality_report.html or .md
- duplicate/near-duplicate report

7.5 Verification checkpoint

PASS when every image has a traceable label and acquisition provenance, class counts are known, and duplicates/near-duplicates are identified.

8. Phase 2 — Leakage-Free Experimental Splits

Objective: make test accuracy trustworthy.

8.1 What not to do

Do not randomly split thousands of patches if those patches came from the same original scan. That can put nearly identical content and fingerprint evidence in both train and test.

8.2 Preferred split unit

Split by document/acquisition session, and where possible evaluate a separate physical device.

Split	Purpose	Rule
Train	Learn model parameters	Only training documents/sessions
Validation	Tune architecture/hyperparameters	Never used for final reporting
Test	Final performance	Locked until experiments are finalized
External/Device test	Generalization	Preferably unseen documents or devices

8.3 Expected output

- train_manifest.csv
- val_manifest.csv
- test_manifest.csv
- split_report.md

8.4 Verification checkpoint

PASS when a script proves there is no overlap of document IDs/session IDs between the relevant splits. If physical-device generalization is claimed, verify device IDs too.

9. Phase 3 — Forensic Preprocessing & Residual Validation

Objective: verify that the residual pipeline is producing meaningful, stable signals.

9.1 Implement/verify

- Grayscale conversion and normalization.
- Consistent resizing/alignment.
- Haar/wavelet residual extraction for offline experiments where appropriate.
- High-pass/Laplacian/KV residual extraction for CNN experiments.
- Document the exact kernel, normalization range, image size, and interpolation method.

9.2 Sanity-check outputs

For a sample image, save:

- original.png

- denoised.png
- residual.png
- residual_histogram.png
- FFT_spectrum.png

9.3 Expected output

A reproducible residual-extraction script plus a folder of visual sanity checks.

9.4 Verification checkpoint

PASS when the residual is numerically stable, has no accidental clipping/overflow, and preprocessing is identical between training and inference.

10. Phase 4 — Traditional ML Baselines

Objective: establish a simple benchmark before evaluating complex models.

10.1 Models

- Random Forest
- RBF SVM

10.2 Features

Use the documented 10-feature baseline only if those features are actually implemented and reproducible. Clearly separate metadata-derived features from forensic residual features.

10.3 Metrics

- Accuracy
- Macro precision
- Macro recall
- Macro F1
- Confusion matrix
- Per-class support

10.4 Expected output

- baseline_metrics.csv
- baseline_confusion_matrix.png
- saved models/scaler/encoder
- baseline_experiment.md

10.5 Verification checkpoint

PASS when rerunning the evaluation with the same frozen test manifest reproduces the reported metrics within a small, documented numerical tolerance.

11. Phase 5 — Deep CNN Baseline (ResNet-18)

Objective: test whether learned residual representations outperform classical features.

11.1 Pipeline

Image → fixed high-pass filter → residual → ResNet-18 → 512-dimensional representation → classifier.

11.2 Record

- Random seed
- optimizer and learning rate
- batch size
- epochs
- scheduler
- weight decay
- augmentation
- early stopping/checkpoint rule
- training/validation curves

11.3 Expected output

- best checkpoint
- training_history.csv
- loss/accuracy curves
- test_metrics.csv
- confusion matrix

11.4 Verification checkpoint

PASS when training and evaluation are reproducible and test performance is computed only after the test set is locked.

12. Phase 6 — Hybrid CNN: Primary Model

Objective: evaluate the central research hypothesis: fusion of learned residual features with explicit forensic descriptors improves attribution.

12.1 Branch A — Residual CNN

256×256×1 residual → fixed Laplacian/high-pass operation → Conv2D 32 → pooling → Conv2D 64 → pooling → Conv2D 128 → global average pooling.

12.2 Branch B — Handcrafted forensic vector

17-dimensional vector containing the implemented PRNU/FFT/LBP/Sobel/statistical measurements.

12.3 Fusion

Concatenate learned and handcrafted representations → Dense 256 → dropout → softmax.

12.4 Expected output

- scanner_hybrid.keras
- feature scaler
- label encoder
- fingerprint database
- training history

- test metrics
- confusion matrix

12.5 Verification checkpoint

PASS when the hybrid model is evaluated on the same locked test manifest as the baselines and its preprocessing/scaler/encoder are identical between training and inference.

13. Phase 7 — Ablation Study

Objective: prove which parts of the proposed method matter.

Experiment	Components
A	CNN only
B	Handcrafted features only + classical classifier
C	PRNU only / PRNU-focused baseline
D	CNN + PRNU
E	CNN + FFT/LBP/statistics
F	CNN + all 17 features (full hybrid)

Run each experiment with the same split and evaluation protocol. Do not change the test set between rows.

Expected output

ablation_results.csv + comparison plot + short interpretation of component contribution.

Verification checkpoint

PASS when the study shows measurable effects of adding/removing components and you can defend the final architecture based on evidence rather than model complexity.

14. Phase 8 — Robustness Testing

Objective: determine whether scanner attribution survives realistic image transformations.

Condition	Test
Original	No transformation
JPEG	Multiple quality levels
Resize	Downscale/upscale
Crop	Remove document margins/regions
Rotation	Small and moderate rotations
Brightness	Moderate changes
Contrast	Moderate changes
Blur	Small blur levels
DPI	Different acquisition resolutions where available

Expected output: robustness_matrix.csv, plots, and examples of failure cases.

Verification checkpoint: PASS when every transformation is generated reproducibly and evaluated against a known source label.

15. Phase 9 — Unknown / Open-Set Scanner Detection

Objective: prevent the system from confidently forcing every unknown image into one of the known classes.

Hold out one scanner/device class during testing. Train on known classes, then evaluate known and held-out sources. Investigate a rejection mechanism using similarity/thresholding or an appropriate open-set method.

Expected output

- known_vs_unknown_metrics.csv
- threshold analysis
- ROC/precision-recall style analysis where appropriate
- unknown examples

Verification checkpoint

PASS when the system has an explicit 'unknown/not sufficiently supported' path and its threshold is chosen using validation data, not the final test set.

16. Phase 10 — Tampering & Patch-Level Anomaly Detection

Objective: determine whether local regions are inconsistent with the document's global scanner fingerprint.

16.1 Pipeline

Document → patch grid → residual extraction → PRNU/feature similarity → anomaly score per patch → heatmap.

16.2 Build controlled tampering data

- Create synthetic edits with known ground-truth locations.
- Record the exact manipulated bounding boxes/masks.
- Keep original and manipulated versions linked by document ID.
- Test multiple manipulation types: copy-paste, text replacement, inserted image, local recompression, and splicing where applicable.

16.3 Metrics

- Patch-level precision/recall/F1
- IoU or overlap with ground-truth masks
- False positive rate
- Localization examples

Expected output

- anomaly heatmaps
- ground-truth masks
- localization_metrics.csv
- false-positive examples

Verification checkpoint

PASS when anomaly performance is measured against known manipulation locations. A visually attractive heatmap alone is not sufficient evidence.

17. Phase 11 — Explainable AI

Objective: provide evidence about what spatial regions influence the CNN decision.

Use Grad-CAM or a comparable method where technically appropriate for the selected architecture.

Expected output: original image + residual + attribution/attention heatmap + predicted class.

Verification checkpoint: PASS when explanations can be generated consistently and are clearly described as model explanations, not causal proof.

18. Phase 12 — Reproducibility & Experiment Management

Create a single experiment configuration and save all parameters with each run.

- Git commit hash
- dataset manifest hash
- Python/dependency versions
- random seeds
- hardware information
- model hyperparameters
- preprocessing version
- training duration
- test metrics
- output artifact paths

Recommended structure:

```
experiments/  
  EXP001_baseline_rf/  
  EXP002_svm/  
  EXP003_resnet18/  
  EXP004_hybrid/  
  EXP005_ablation_prnu/  
  EXP006_robustness/  
  EXP007_open_set/  
  EXP008_tampering/
```

Expected output: a results registry such as experiments.csv plus saved configs and logs.

Verification checkpoint: another clean environment should be able to reproduce the key evaluation pipeline from the saved manifests/configuration.

19. Phase 13 — Final Research Evaluation

Before writing the paper, freeze the methodology and prepare one master result table.

Method	Accuracy	Macro F1	Inference time	Unknown handling	Notes
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Random Forest	Actual result	Actual result	Measured	No/Yes	Baseline
SVM	Actual result	Actual result	Measured	No/Yes	Baseline
ResNet-18	Actual result	Actual result	Measured	No/Yes	Deep baseline
Hybrid CNN	Actual result	Actual result	Measured	No/Yes	Proposed model

Do not fill the table with target numbers. Use only measurements from the locked evaluation protocol.

20. Research Paper Structure

1. Title — precise and non-sensational.
2. Abstract — problem, method, dataset, evaluation, main measured result, limitations.
3. Introduction — motivation and research gap.
4. Related Work — scanner attribution, PRNU, image forensics, CNN-based source attribution, document tampering.
5. Research Questions / Contributions — explicitly state what is being investigated.
6. Dataset and Experimental Protocol — acquisition, classes, devices, documents, splits, preprocessing.
7. Methodology — residual extraction, PRNU, handcrafted features, CNN, fusion, classifier.
8. Experiments — baselines, ablations, robustness, open-set, tampering.
9. Results — tables, confusion matrices, statistical/uncertainty discussion where appropriate.
10. Discussion — why the model behaves as observed and where it fails.
11. Limitations — dataset size, device diversity, transformations, scanner models, forensic validation limits.
12. Conclusion — what the experiments establish and what remains future work.
13. References — primary research papers and authoritative technical sources.

21. Final Go/No-Go Checklist

Question	PASS condition
Is the actual implementation verified?	Every major claim maps to code/artifact.
Is the dataset documented?	Exact image/document/device counts are known.
Is leakage prevented?	No prohibited overlap across evaluation splits.
Is the test set locked?	Final metrics come from a fixed independent test manifest.
Are baselines reproducible?	RF/SVM results can be regenerated.
Is the CNN reproducible?	ResNet training/evaluation configuration is saved.
Is the hybrid gain demonstrated?	Ablation study compares the same evaluation protocol.
Is robustness measured?	Transformations and performance are documented.
Is unknown-source behavior tested?	Known vs unknown evaluation exists if claimed.
Is tampering evaluated quantitatively?	Ground truth masks and localization metrics exist.
Are claims appropriately scoped?	Model confidence is not presented as legal proof.
Can another person reproduce the main result?	Code + manifests + config + dependencies are available.

22. Suggested Execution Order

Do not attempt all upgrades simultaneously. Work in this order:

14. Audit the repository.
15. Freeze and clean the dataset.
16. Create leakage-free manifests.
17. Validate residual extraction visually and numerically.

18. Reproduce Random Forest and SVM baselines.
19. Reproduce ResNet-18.
20. Reproduce the Hybrid CNN.
21. Run the ablation study.
22. Run robustness tests.
23. Add open-set scanner detection.
24. Build and quantitatively evaluate tampering localization.
25. Add explainability.
26. Freeze all experiments and generate the master results table.
27. Write the research paper from the measured evidence.

23. Verification Report Template — Use After Every Phase

Copy this template into docs/phase_reports/ after each completed phase.

Field	What to record
Phase	Phase number and name
Objective	What this phase was supposed to prove/build
Changes made	Files/code/configuration changed
Input	Dataset/model/config used
Output	Exact files, metrics, plots, models
Expected result	What should happen
Actual result	What actually happened
Verification	PASS / FAIL / NEEDS REVIEW
Problems	Errors, leakage, instability, limitations
Next action	Exact work required before next phase

24. First Deliverable to Create Now

Start with Phase 0. Do not modify the model yet. First produce an implementation audit and a dataset inventory. This establishes the factual baseline for everything that follows.

Minimum files after Phase 0:

- docs/implementation_audit.md
- docs/dataset_inventory.md
- docs/current_results.md
- docs/known_limitations.md

Only after these are complete should you begin Phase 1. The purpose is to prevent building a research paper around features that are documented but not actually implemented or around metrics produced by a potentially leaky split.

25. End Goal

The final TraceScope AI research system should not merely contain more algorithms. It should provide a defensible experimental chain: controlled data → leakage-free split → validated residual extraction →

reproducible baselines → proposed hybrid model → ablation → robustness → unknown-source testing → quantitative tampering localization → reproducible results → carefully scoped research claims.

When all checkpoints pass, the project will have a much stronger foundation for a research paper because every major claim can be traced to a dataset, experiment, metric, and reproducible output.